

MATH403: Abstract Algebra

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§1 Preliminaries

Theorem 1.1

Well Ordering Principle states that every nonempty set of positive integers contains a smallest number.

Definition 1.2. A nonzero integer t is a **divisor** of an integer s if there is an integer u such that $s = tu$ and we write $t|s$. A **prime** is a positive integer greater than 1 whose only positive divisors are 1 and itself. We say an integer s is a **multiple** of an integer t if there is an integer u such that $s = tu$.

Theorem 1.3

Division Algorithm: let a and b be integers with $b > 0$. Then there exist unique integers q and r with the property that $a = bq + r$, where $0 \leq r < b$.

Proof. Omitted. □

Definition 1.4. The **greatest** common divisor of two integers is the largest of all common divisors of a and b . We denote this by $\gcd(a, b)$. When $\gcd(a, b) = 1$, we say a and b are **relatively prime**.

Theorem 1.5

For any nonzero integers a and b , there exist integers s and t such that $\gcd(a, b) = as + bt$. Moreover, $\gcd(a, b)$ is the smallest positive integer of the form $as + bt$.

Proof. Omitted. □

Corollary 1.6

Integers a and b are relatively prime if and only if there exist integers s and t such that $as + bt = 1$.

Lemma 1.7 (Euclid's Lemma)

If p is a prime that divides ab , then p divides a or p divides b .

Proof. Omitted. □

Theorem 1.8 (Fundamental Theorem of Arithmetic)

Every integer greater than 1 is a prime or a product of primes. This product is unique, except for the order in which factors appear.

Definition 1.9. The **least common multiple** of two nonzero integers a and b is the smallest positive integer that is a multiple of both a and b . We denote this as $\text{lcm}(a, b)$.

Definition 1.10 (Modular Arithmetic). When $a = qn + r$ where q is the quotient and r is the remainder upon dividing a by n , we write $a \bmod n = r$.

Definition 1.11 (Equivalence Relation). A **equivalence relation** on a set S is a set R of ordered pairs of elements of S such that

1. Reflexive: $(a, a) \in R$ for all $a \in S$
2. Symmetric: $(a, b) \in R \implies (b, a) \in R$
3. Transitive: $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$

If $\sim$ is an equivalence relation on a set S and $a \in S$, then the set $[a] = \{x \in S \mid x \sim a\}$ is called the **equivalence class** of S containing a .

Definition 1.12. A **partition** of a set S is a collection of nonempty disjoint subsets of S whose union is S .

Theorem 1.13

The equivalence classes of an equivalence relation on a set S constitute a partition of S . Conversely, for any partition P of S , there is an equivalence relation on S whose equivalence classes are the elements of P .

Proof. Omitted □

Definition 1.14. A function ϕ from a set A to set B is a rule that assigns to each element a of A exactly one element b of B . A is the **domain**, B is the **range**, and the subset of B comprising all images of elements of A is called the **image of A under ϕ** .

Definition 1.15. Let $\phi : A \rightarrow B$ and $\varphi : B \rightarrow C$. The **composition** $\varphi\phi$ is the mapping from A to C defined by $\varphi(\phi(a))$ for all a in A .

Definition 1.16. A function ϕ from a set A is **one-to-one** if for every $a_1, a_2 \in A$ then

$$\phi(a_1) = \phi(a_2) \implies a_1 = a_2$$

Definition 1.17. A function $\phi : A \rightarrow B$ is **onto** if each element of B is the image of at least one element of A .

$$\forall b \in B, \exists a \in A \text{ s.t. } \phi(a) = b$$

Theorem 1.18

Given functions $\alpha : A \rightarrow B, \beta : B \rightarrow C, \gamma : C \rightarrow D$ then

- Associativity: $\gamma(\beta\alpha) = (\gamma\beta)\alpha$
- If α, β are both 1-1, then $\beta\alpha$ is 1-1
- If α, β are onto, then $\beta\alpha$ is onto
- If α is 1-1 and onto, then there is a function α^{-1} from B onto A such that $(\alpha^{-1}\alpha)(a) = a \forall a \in A$ and $(\alpha\alpha^{-1})(b) = b \forall b \in B$

§2 Introduction to Groups Through Dihedral Groups

A solid introduction to groups is considering the symmetries of a square

	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_0	R_0	R_{90}	R_{180}	R_{270}	H	V	D	D'
R_{90}	R_{90}	R_{180}	R_{270}	R_0	D'	D	H	V
R_{180}	R_{180}	R_{270}	R_0	R_{90}	V	H	D'	D
R_{270}	R_{270}	R_0	R_{90}	R_{180}	D	D'	V	H
H	H	$\mathbb{D}$	V	D'	R_0	R_{180}	R_{90}	R_{270}
V	V	D'	H	D	R_{180}	R_0	R_{270}	R_{90}
D	D	V	D'	H	R_{270}	R_{90}	R_0	R_{180}
D'	D'	H	D	V	R_{90}	R_{270}	R_{180}	R_0

The above is the Cayley table for the **dihedral group of order 8**, which is denoted by D_4 . There are some key properties of this group that we will formalize later.

Definition 2.1. The idea above for a square can be done for any regular n -gon ($n \geq 3$), and the corresponding group is denoted by D_n and is called the **dihedral group of order $2n$** .

Definition 2.2. Let G be a set. The **binary operation** on G is a function that assigns each ordered pair of elements of G an element of G . This condition is called **closure**.

Definition 2.3. Let G be a set together with a binary operation that assigns to each ordered pair of elements of G an element G denoted by ab . We say G is a **group** under this operation if the following three properties are satisfied

1. Closure?
2. Associativity: $(ab)c = a(bc) \forall a, b, c \in G$
3. Identity: There is an element e called the identity in G such that $ae = ea = a \forall a \in G$

4. Inverses: for each element $a \in G$, there is an element b in G such that $ab = ba = e$.

Definition 2.4. If a group has the property that $ab = ba \forall a, b \in G$, then the group is **Abelian**.

Example 2.5 • $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ are all groups under ordinary addition. In each case, the identity is 0 and the inverse of a is $-a$.

- The set of integers under ordinary multiplication is not a group, it fails the inverse property check.
- The set $Z_n = \{0, 1, \dots, n-1\}$ for $n \geq 1$ is a group under addition modulo n .
- The set $\mathbf{R}^*$ of nonzero real numbers is a group under ordinary multiplication with identity 1 and inverse $\frac{1}{a}$ for any a .
- The set

$$GL(2, \mathbb{R}) = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mid a, b, c, d \in \mathbb{R}, ad - bc \neq 0 \right\}$$

is a non-Abelian group under the operation of matrix multiplication.

Theorem 2.6 (Uniqueness of the Identity)

In a group G , there is only one identity element.

Proof. Omitted. □

Theorem 2.7 (Cancellation)

In a group G , the right and left cancellation laws hold; that is, $ba = ca \implies b = c$ and $ab = ac \implies b = c$.

Proof. Omitted. □

Theorem 2.8 (Uniqueness of Inverses)

For each element $a \in G$, there is a unique element $b \in G$ such that $ab = ba = e$.

Proof. Omitted. □

Theorem 2.9 (Socks-Shoes Property)

For $a, b \in G$, $(ab)^{-1} = b^{-1}a^{-1}$

Proof. Omitted. □

§3 Finite Groups and Subgroups

Groups with finitely many elements have interesting arithmetic properties.

Definition 3.1 (Order of a Group). The number of elements in a group (finite or infinite) is called its **order**. We will use $|G|$ to denote the order of G .

Example 3.2

The group $\mathbb{Z}$ under addition has infinite order, whereas the group $U(1, 3, 7, 9)$ under multiplication module 10 has order 4.

Definition 3.3. The order g in a group G is the smallest positive integer n such that $g^n = e$. In additive notation, this would be $ng = e$. If no such integer exists, we say that g has infinite order. The order of an element g is denoted by $|g|$.

Note 3.4. The group $U(n)$ is the set of all positive integers less than and relatively prime to n .

Definition 3.5 (Subgroup). If a subset H of a group is itself a group under the operation of G , we say that H is a subgroup of G . Subgroups still must have the same properties as groups (closure, associativity, identity, and inverses).

§3.1 Subgroup Tests

Theorem 3.6 (One-Step Subgroup Test)

Let G be a group and H a nonempty subset of G . If ab^{-1} is in H whenever $a, b \in H$, then H is a subgroup of G . In additive notation, if $a - b \in H$ whenever $a, b \in H$, then H is a subgroup of G .

Proof. Omitted. □

There are actually four steps involved in applying the theorem.

1. Identify the property P that distinguishes the elements of H ; that is, identify a defining condition.
2. Prove that the identity has property P ; this verifies that H is nonempty
3. Assume that two elements a and b have property P .
4. Use the assumption that a and b have property P to show that ab^{-1} has property P .

Theorem 3.7 (Two-Step Subgroup Test)

Let G be a group and let H be a nonempty subset of G . If $ab \in H$ whenever $a, b \in H$ and $a \in H \implies a^{-1} \in H$, then H is a subgroup of G .

- H is closed under the operation, $a, b \in H \implies ab \in H$
- H is closed under taking inverses, $a \in H \implies a^{-1} \in H$

Proof. Omitted. □

Same process as above, but we use the assumption that a and b have property P to prove that ab has property P and a^{-1} has property P .

Note 3.8. How do you show that a subset of a group is *not* a subgroup?

1. Show that the identity is not in the set.
2. Exhibit an element of the set whose inverse is not in the set.
3. Exhibit two elements of the set whose product is not in the set.

Theorem 3.9 (Finite Subgroup Test)

Let H be a nonempty finite subset of a group G . If H is closed under the operation of G , then H is a subgroup of G .

Definition 3.10. For any element $a \in G$, we let $\langle a \rangle = \{a^n \mid n \in \mathbb{Z}\}$. In particular, observe that the exponents of a include all negative integers as well as 0 and the positive integers, a^0 is defined to be the identity

Theorem 3.11 ($\langle a \rangle$ is a Subgroup)

Let G be a group, and let $a \in G$. Then $\langle a \rangle$ is a subgroup of G .

Proof. Since $a \in \langle a \rangle$, $\langle a \rangle$ is not empty. Let $a^n, a^m \in \langle a \rangle$. Then $a^n(a^m)^{-1} = a^{n-m} \in \langle a \rangle$; so by the One-Step Subgroup Test, $\langle a \rangle$ is a subgroup of G . □

Definition 3.12 (Center of a Group). The **center**, $Z(G)$, of a group G is the subset of elements in G that commute with every element of G .

$$Z(G) = \{a \in G \mid ax = xa \forall x \in G\}$$

Theorem 3.13 (Center is a Subgroup)

The center of a group G is a subgroup of G .

Proof. Omitted. □

Definition 3.14 (Centralizer of a in G). Let a be a fixed element of G . The **centralizer** of a in G , $C(a)$, is the set of all elements in G that commute with a

$$C(a) = \{g \in G \mid ga = ag\}$$

Example 3.15

In D_4 , we have the following centralizers:

- $C(R_0) = D_4 = C(R_{180})$
- $\vdots$

Theorem 3.16 ($C(a)$ is a Subgroup)

For each $a \in G$, the centralier of a is a subgroup of G .

Proof. Omitted. □

§4 Cyclic Groups

Definition 4.1. A group G is **cyclic** if there is an element $a \in G$ such that $G = \{a^n \mid n \in \mathbb{Z}\}$. Such an element is called a **generator** of G . From now on, we may indicate that G is a cyclic group generated by a by writing $G = \langle a \rangle$.

Theorem 4.2 (Criterion for $a^i = a^j$)

Let G be a group, and let $a \in G$.

- If a has infinite order, then $a^i = a^j$ if and only if $i = j$
- If a has finite order, then $a^i = a^j$ if and only if n divides $i - j$.

Corollary 4.3

For any group element $a \in G$, $|a| = |\langle a \rangle|$.

Corollary 4.4

For any group element a , $a^k = e$ if and only if $|a|$ divides k .

Corollary 4.5

If a, b belong to a finite group and $ab = ba$ then $|ab|$ divides $|a||b|$.

Theorem 4.6

Let $a \in G$ be such that $|a| = n$, and let k be a positive integer. Then $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$ and $|a^k| = \frac{n}{\gcd(n,k)}$.

Corollary 4.7

In a finite cyclic group, the order of an element divides the order of the group.

Corollary 4.8

Let $|a| = n$. Then $\langle a^i \rangle = \langle a^j \rangle$ if and only if $\gcd(n, i) = \gcd(n, j)$ and $|a^i| = |a^j|$ if and only if $\gcd(n, i) = \gcd(n, j)$.

Corollary 4.9

Let $|a| = n$. Then $\langle a \rangle = \langle a^j \rangle$ if and only if $\gcd(n, j) = 1$.

Corollary 4.10

An integer $k \in Z_n$ is a generator of Z_n if and only if $\gcd(n, k) = 1$.

Theorem 4.11

Every subgroup of a cyclic group is cyclic. Moreover, if $|\langle a \rangle| = n$, then the order of any subgroup of $\langle a \rangle$ is a divisor of n ; for each positive integer k on n , the group $\langle a \rangle$ has exactly one subgroup of order k — $\langle a^{n/k} \rangle$

Corollary 4.12

For each divisor k of n , the set $\langle n/k \rangle$ is the unique subgroup of Z_n of order k ; moreover, these are the only subgroups of Z_n .

■ **EXAMPLE 7** The list of subgroups of Z_{30} is

$\langle 1 \rangle$	$= \{0, 1, 2, \dots, 29\}$	order 30,
$\langle 2 \rangle$	$= \{0, 2, 4, \dots, 28\}$	order 15,
$\langle 3 \rangle$	$= \{0, 3, 6, \dots, 27\}$	order 10,
$\langle 5 \rangle$	$= \{0, 5, 10, 15, 20, 25\}$	order 6,
$\langle 6 \rangle$	$= \{0, 6, 12, 18, 24\}$	order 5,
$\langle 10 \rangle$	$= \{0, 10, 20\}$	order 3,
$\langle 15 \rangle$	$= \{0, 15\}$	order 2,
$\langle 30 \rangle$	$= \{0\}$	order 1.

■

Theorem 4.13 (Number of Elements of Each Order in a Cyclic Group)

Corollary 4.14 (Number of Elements of Order d in a Finite Group)

In a finite group, the number of elements of order d is a multiple of $\phi(d)$, where ϕ is the **Euler Phi** function, which returns all of the positive integers less than n and relatively prime to n , and $\phi(1) = 1$.

§5 Permutation Groups

At this point, we know two families of finite Abelian groups $Z_n = \{0, 1, \dots, n-1\}$ and $U(n)$ the set of all positive integers less than n that are relatively prime with n . We also know two families of infinite non-Abelian groups, dihedral groups and matrix groups.

Definition 5.1 (Permutation of A , Permutation Group of A). A **permutation** of a set A is a function from A to A that is both 1-1 and onto. A **permutation group** of a set A is a set of permutations of A that forms a group under function composition.

Definition 5.2. **Cycle notation** has theoretical advantages in that certain important properties of the permutation can be readily determined when cycle notation is

used. Consider the permutation

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 4 & 6 & 5 & 3 \end{bmatrix}$$

Instead, we have $\alpha = (12)(346)(5)$, where for example, (346) is a 3-cycle.

Theorem 5.3 (Properties of Disjoint Cycles)

Every permutation of a finite set can be written as a cycle or as a product of disjoint cycles.

Theorem 5.4 (Disjoint Cycles Commute)

If the pair of cycles $\alpha = (a_1, a_2, \dots, a_m)$ and $\beta = (b_1, b_2, \dots, b_n)$ have no entries in common, then $\alpha\beta = \beta\alpha$.

Theorem 5.5 (Order of a Permutation)

The order of a permutation of a finite set written in disjoint cycle form is the least common multiple of the lengths of the cycles.

Theorem 5.6 (Product of 2-Cycles)

Every permutation in $S_n, n > 1$ is a product of 2-cycles.

Theorem 5.7

If $e = \beta_1\beta_2 \dots \beta_r$ where the β 's are 2-cycles, then r is even.

Theorem 5.8 (Always Even or Always Odd)

If a permutation α can be expressed as a product of an even (odd) number of 2-cycles, then every decomposition of α into a product of 2-cycles must have an even (odd) number of 2-cycles.

Definition 5.9 (Even and Odd Permutations). A permutation that can be expressed as a product of an even (odd) number of 2-cycles is called an **even (odd)** permutation.

Theorem 5.10

The set of even permutations in S_n forms a subgroup of S_n .

Definition 5.11. The group of even permutations of n symbols is denoted by A_n and is called the **alternating group of degree n** .

Theorem 5.12

For $n > 1$, A_n has order $n!/2$.

§6 Isomorphisms

Definition 6.1. An **isomorphism** ϕ from a group G to a group G' is a 1-1 and onto mapping from G to G' that preserves the group operation. That is,

$$\phi(ab) = \phi(a)\phi(b) \quad \forall a, b \in G$$

If there is an isomorphism from G onto G' , then we say that G and G' are isomorphic and we write $G \approx G'$.

G	—	—	—	b	—	—
—	—	—	—	—	—	—
—	—	—	—	—	—	—
a	—	—	—	ab	—	—
—	—	—	—	—	—	—
$\overline{G}$	—	—	—	$\phi(b)$	—	—
—	—	—	—	—	—	—
—	—	—	—	—	—	—
$\phi(a)$	—	—	—	$\phi(ab)$	—	—
—	—	—	—	—	—	—

Table 6.1

G Operation	$\overline{G}$ Operation	Operation Preservation
$\cdot$	$\cdot$	$\phi(a \cdot b) = \phi(a) \cdot \phi(b)$
$\cdot$	$+$	$\phi(a \cdot b) = \phi(a) + \phi(b)$
$+$	$\cdot$	$\phi(a + b) = \phi(a) \cdot \phi(b)$
$+$	$+$	$\phi(a + b) = \phi(a) + \phi(b)$

Note 6.2. To prove that a group G is isomorphic to a group G' ,

1. Define a function $\phi : G \rightarrow G'$
2. 1-1: prove that ϕ is 1-1, so $\phi(a) = \phi(b) \implies a = b$
3. Onto: prove that ϕ is onto, so for any $g' \in G'$, find a $g \in G$ such that $\phi(g) = g'$
4. Prove that ϕ is operation-preserving; that is show that $\phi(ab) = \phi(a)\phi(b) \quad \forall a, b \in G$.

Theorem 6.3 (Properties of Isomorphisms Acting on Elements)

1. ϕ carries the identity of G to the identity of G'
2. For every integer n and for every $a \in G$, then $\phi(a^n) = [\phi(a)]^n$, and the additive form is $\phi(na) = n\phi(a)$
3. For any $a, b \in G$, a and b commute if and only if $\phi(a)$ and $\phi(b)$ commute
4. $G = \langle a \rangle$ iff $G' = \langle \phi(a) \rangle$
5. $|a| = |\phi(a)| \ \forall a \in G$ — isomorphisms preserve orders
6. If G is finite, then G and G' have exactly the same number of elements of every order.

Theorem 6.4 (Properties of Isomorphisms Acting on Groups)

1. ϕ^{-1} is an isomorphism from G' onto G
2. G is Abelian if and only if G' is Abelian
3. G is cyclic if and only if G' is cyclic
4. If K is a subgroup of G , then $\phi(K) = \{\phi(k) \mid k \in K\}$ is a subgroup of G'
5. If K' is a subgroup of G' , then $\phi^{-1}(K') = \{g \in G \mid \phi(g) \in K'\}$ is a subgroup of G
6. $\phi(Z(G)) = Z(G')$

§6.1 Automorphisms

Definition 6.5. An isomorphism from a group G onto itself is called an **automorphism** of G .

Definition 6.6. Let G be a group, and let $a \in G$. The function ϕ_a defined by $\phi_a(x) = axa^{-1}$ for all $x \in G$ is called the **inner automorphism of G induced by a** .

Theorem 6.7

The set of automorphisms of a group and the set of inner automorphisms of a group are both groups under the operation of function composition.

Theorem 6.8 (Cayley's Theorem)

Every group is isomorphic to a group of permutations.

§7 Cosets and Lagrange's Theorem

Definition 7.1. Let G be a group and let H be a nonempty subset of G . For any $a \in G$, the set

$$aH = \{ah \mid h \in H\} \quad \{Ha = \{ha \mid h \in H\}\}$$

When H is a subgroup of G , the set aH is called the **left coset of H in G containing a** , and similarly for **right coset**. In this case, the element a is the **coset representative**.

Lemma 7.2

Let H be a subgroup of G , and let $a, b \in G$. Then

1. $a \in aH$
2. $aH = H$ if and only if $a \in H$
3. $(ab)H = a(bH)$ if and only if $H(ab) = (Ha)b$
4. $aH = bH$ if and only if $a \in bH$
5. $aH = bH$ or $aH \cap bH = \emptyset$
6. $aH = bH$ if and only if $a^{-1}b \in H$
7. $|aH| = |bH|$

Theorem 7.3 (Lagrange's Theorem)

If G is a finite group and H is a subgroup of G , then $|H|$ divides $|G|$. Moreover, the number of distinct left (right) cosets of H in G is $|G|/|H|$.

Corollary 7.4

In a finite group, the order of each element of the group divides the order of the group.

Corollary 7.5

Every group of prime order is isomorphic to Z_p .

Corollary 7.6

Let G be a finite group, and let $a \in G$. Then, $a^G = e$.

Corollary 7.7

For every integer a and every prime p , $a^p \bmod p = a \bmod p$.

Theorem 7.8

For two finite subgroups H and K of a group, define the set $HK = \{hk \mid h \in H, k \in K\}$. Then

$$|HK| = |H||K| = |H||K|/|H \cap K|$$

Theorem 7.9

Let G be a group of order $2p$, where p is a prime greater than 2. Then G is isomorphic to Z_{2p} or D_p .

§8 External Direct Products

Definition 8.1 (External Direct Product). Let $G_1, \dots, G_n$ be a finite collection of groups. The external direct product of such groups written as $G_1 \oplus G_2 \oplus \dots \oplus G_n$ is the set of all n -tuples for which the i -th component is an element of G_i , and the operation is componentwise.

Example 8.2

Indeed, $\mathbb{R}^2 = \mathbb{R} \oplus \mathbb{R}$ and the operation is componentwise/vector addition. Another example is

$$Z_2 \oplus Z_3 = \{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)\}$$

Theorem 8.3

The order of an element in a direct product of a finite number of finite groups is the LCM of the orders of the components of the element.

$$|(g_1, g_2, \dots, g_n)| = \text{lcm}(|g_1|, |g_2|, \dots, |g_n|)$$

Theorem 8.4 (Criterion for $G \oplus H$ to be Cyclic)

Let G and H be finite cyclic groups. Then $G \oplus H$ is cyclic if and only if $|G|$ and $|H|$ are relatively prime.

Corollary 8.5 (Criterion for $G_1 \oplus G_2 \oplus \dots \oplus G_n$ to be cyclic)

An external direct product $G_1 \oplus G_2 \oplus \dots \oplus G_n$ of a finite number of finite cyclic groups is cyclic if and only if $|G_i|$ and $|G_j|$ are relatively prime when $i \neq j$.

Corollary 8.6

Let $m = n_1 n_2 \dots n_k$. Then Z_m is isomorphic to $Z_{n_1} \oplus Z_{n_2} \oplus \dots \oplus Z_{n_k}$ if and only if n_i and n_j are relatively prime when $i \neq j$.

Example 8.7

For example, we have

$$Z_2 \oplus Z_2 \oplus Z_3 \oplus Z_5 \approx Z_2 \oplus Z_6 \oplus Z_5 \approx Z_2 \oplus Z_{30}$$

but note that $Z_2 \oplus Z_{30} \not\approx Z_{60}$

Theorem 8.8 ($U(n)$ as an External Direct Product)

Suppose s and t are relatively prime. Then $U(st)$ is isomorphic to the external direct product of $U(s)$ and $U(t)$. In short,

$$U(st) \approx U(s) \oplus U(t)$$

Moreover, $U_s(st)$ is isomorphic to $U(t)$ and $U_t(st)$ is isomorphic to $U(s)$.

Corollary 8.9

Let $m = n_1 n_2 \dots n_k$ where $\gcd(n_i, n_j) = 1 \forall i \neq j$. Then

$$U(m) \approx U(n_1) \oplus U(n_2) \oplus \dots \oplus U(n_k)$$

§9 Normal Subgroups and Factor Groups

Definition 9.1 (Normal Subgroup). A subgroup H of a group G is called a **normal subgroup** of G if $aH = Ha \forall a \in G$. We denote this by $H \triangleleft G$.

Theorem 9.2 (Normal Subgroup Test)

A subgroup H of G is normal in G if and only if $xHx^{-1} \subseteq H \forall x \in G$.

Theorem 9.3 (Factor Groups)

Let G be a group and let H be a normal subgroup of G . The set $G/H = \{aH \mid a \in G\}$ is a group under the operation $(aH)(bH) = abH$.

Theorem 9.4 (G/Z Theorem)

Let G be a group and let $Z(G)$ be the center of G . If $G/Z(G)$ is cyclic, then G is Abelian.

Theorem 9.5 ($G/Z(G) \approx \text{Inn}(G)$)

For any group G , $G/Z(G) \approx \text{Inn}(G)$

Theorem 9.6